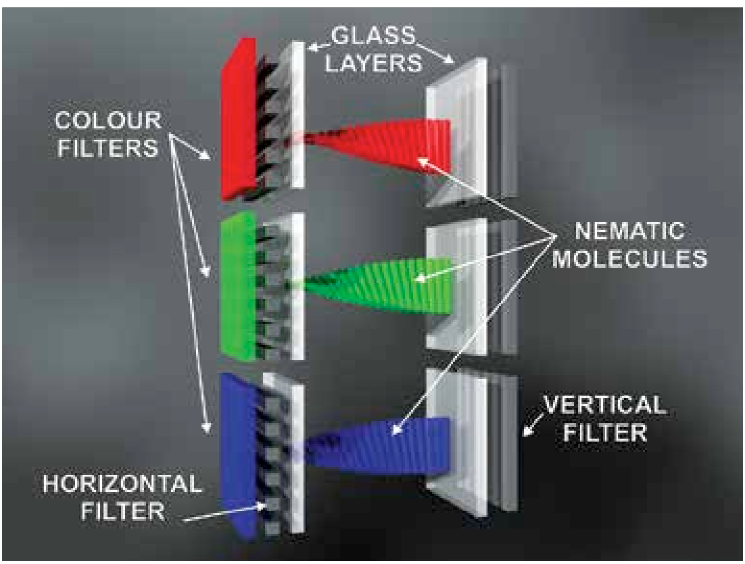
On an environmental level, CRTs are much more harmful than TN LCD displays, due to the comparatively more radiation emitted and almost 200 percent electricity consumed as compared to LCDs. Even the display quality is prey to glare and lack of sharpness for CRT displays whereas both effects are barely felt in LCDs, not to mention the immense heating problem.

How does a TN display work?

The underlying principle in a TN display is that of polarized light. Any display is made of pixels. In case of an LCD display, each of these pixels is typically a liquid molecule layer between two transparent electrodes and two polarizing light filters set perpendicular to each other. In its uncharged state, the light incoming through the first filter cannot pass through the second filter due to the different orientations of the polarizing filter. When a low voltage is applied to the molecules, the orientation is changed to bend the light between the two filters to match the orientation of the second filter as well. If the orientation is completely changed then the light is passed unfiltered. By varying the voltage applied, various levels of intensity can be achieved.



A small section of a typical color TN panel. Notice the nematic molecules which bend the light between the filters.